

EDUCATION

Indiana University Bloomington (IUB)

Ph.D. in Intelligent Systems Engineering, GPA: 3.8

Bloomington, IN

Aug 2024–Current

- Track: Computer Engineering
- Minor: Computer Science
- Related Coursework:

- HPC
- Graph Analytics
- Deep Learning Systems
- Distributed Systems
- Engineering Compilers
- ML Signal Processing

Grand Valley State University (GVSU)

B.S. in Computer Science, GPA: 3.9

Grand Rapids, MI

Aug 2020–Apr 2024

- Minor: Mathematics
- Related Coursework:

- Theory of Computation
- Linear Algebra II
- Applied Machine Learning
- Numerical Analysis
- Algorithms Engineering
- Scientific Computing

PROFESSIONAL EXPERIENCES

Graduate Research Assistant

Indiana University

Advisor: Dr. Fengguang Song

Error-Bound Lossy Compression for Scientific Data (FZ)

Aug 2024–Current

- FZModules: A modular compression framework for heterogeneous error-bounded lossy compression of scientific data. <https://github.com/szcompressor/FZModules>. Primary developer.
- Working on lossy error-bounded compression for scientific applications such as Advanced Photon Source data, creating tooling the further adoption of FZ compressors, and optimizing high-performance data reduction modules.

Graduate Teaching Assistant

Indiana University

Cloud Computing

Aug 2025–Dec 2025

- Teaching assistant for graduate level cloud computing course. Worked with students on lab assignments and the course project.

Summer Undergraduate Research Internship

Sandia National Labs

Advisor: Dr. Oksana Guba

Optimizing the E3SM Climate Model

May 2024–Aug 2024

- Worked towards understanding redundant solves in the world class E3SM climate model, helping maximize its potential for exascale applications.

Undergraduate Researcher

Applied Computing Institute (GVSU)

Advisors: Dr. Zachary DeBruine & Dr. Erin Carrier

IVSparse - Sparse Data Compression Library

Aug 2022–May 2024

- Created two proprietary compression formats which leverage present redundancy in data to allow for usable compressed data with a limited performance loss.
- Involved work in high performance computing (HPC), data compression, and data structure design.

PUBLICATIONS

- **Best Short Paper Award** **Skyler Ruiter**, Jiannan Tian, Fengguang Song. “FZModules: A Heterogeneous Computing Framework for Customizable Scientific Data Compression Pipelines.” *2025 IEEE/ACM The 11th International Workshop on Data Analysis and Reduction for Big Scientific Data (DRBSD)*. St Louis, MO, November 16-21. (Accepted)
- Seth Wolfgang, **Skyler Ruiter**, Marc Tunnell, Timothy Triche Jr, Erin Carrier, Zachary DeBruine. “Value-Compressed Sparse Column (VCSC): Sparse Matrix Storage for Single-cell Omics Data.” *2024 IEEE International Conference on Big Data (BigData)*. Washington D.C., December 15-18.

TALKS

- (FZModules) ZF/FZ Joint Workshop Washington DC., October 2025
- (FZModules) FZ/ZF Joint Workshop Sarasota FL, March 2025
- (IVSparse) IEEE BigData HPC-BOD Workshop Presentation Washington DC., December 2024
- (IVSparse) IEEE Data Compression Conference Technical Presentation Salt Lake City UT, March 2024
- (IVSparse) GVSU School of Computing Seminar Series Allendale MI, February 2024
- (IVSparse/Genomic Modeling) GVSU Undergraduate Research Fair Allendale MI, October 2023
- (IVSparse) Grand Rapids Tech Week: Engineering and Computing Showcase Grand Rapids MI, September 2023
- (IVSparse) Grand Valley State University Student Scholar’s Day Allendale MI, April 2023

EXTRACURRICULAR ACTIVITIES

GVSU Computing Club President Spring 2021–Spring 2024

- Networked with companies, brought in many professors and researchers, and ran events and meetings regularly.
- Mentored numerous other undergraduate students, helping achieve internships, begin research positions, and plan their educational and career goals.

Dean’s Student Advisory Council Spring 2022–Spring 2024

- Advised the dean of the Padnos College of Engineering and Computing with a small collection of other students from different disciplines.

SCHOLARSHIPS AND AWARDS

- Luddy Doctoral AI Fellowship Fall 2024
- Science Undergraduate Laboratory Internships (SULI) Award Summer 2024
- P. Douglas Kindschi Undergraduate Research Fellowship Spring 2024

RESEARCH INTERESTS

- High Performance Computing
- Data Compression
- Scientific Computing
- Sparse Data Analysis

TOOLING

- Languages/Libraries: C++, Python, PyTorch, Rust
- GPU Programming: CUDA, NCCL
- HPC Tooling: OpenMP, MPI, CMake
- Performance Portability: Kokkos, CUDASTF